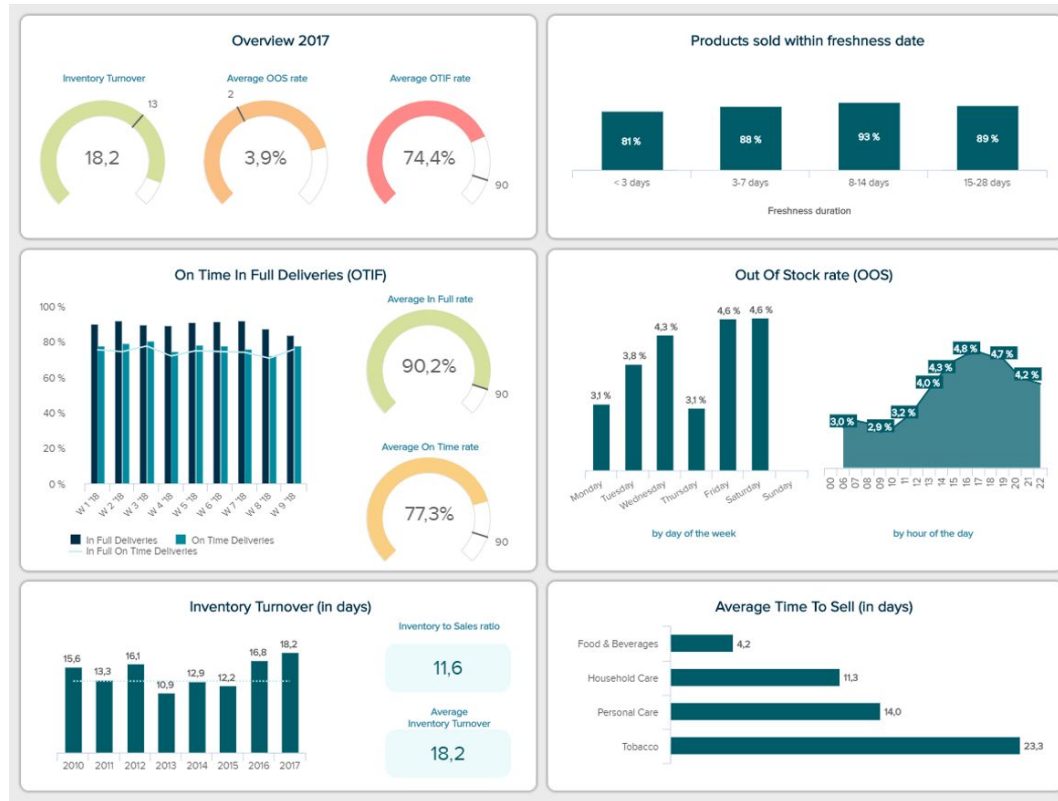




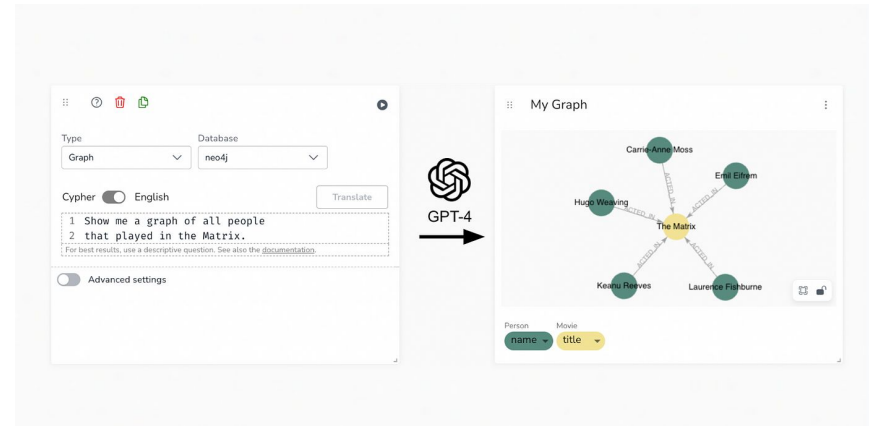
Developing Analysis Dashboards using LLMs

Introduction



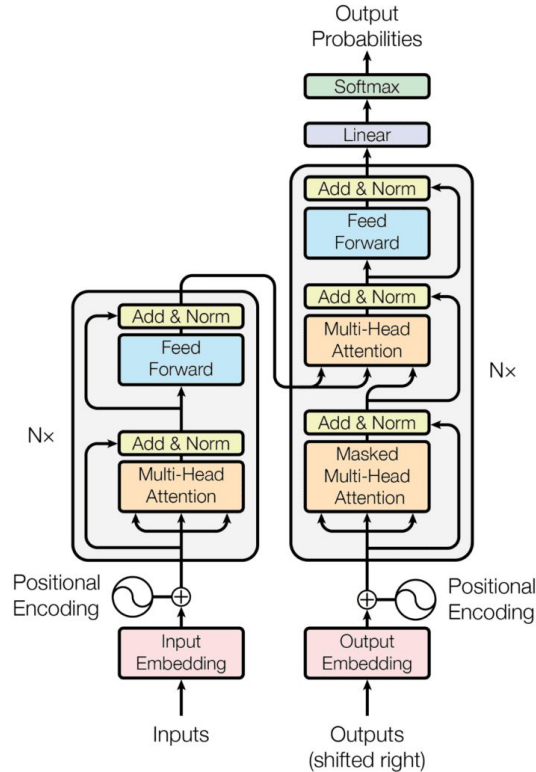
Why use LLMs for Analytics?

- Enhanced Data Exploration
- Improved Decision-Making
- Increased Accessibility
- Efficient Knowledge Discovery
- Continuous Learning and Improvement
- Improve Productivity



Large Language Models

From Transformer Model to Large Language Model



What are some use cases for LLMs?

- Text Generation
- Language Translation
- Text Summarization
- Question Answering

Langchain

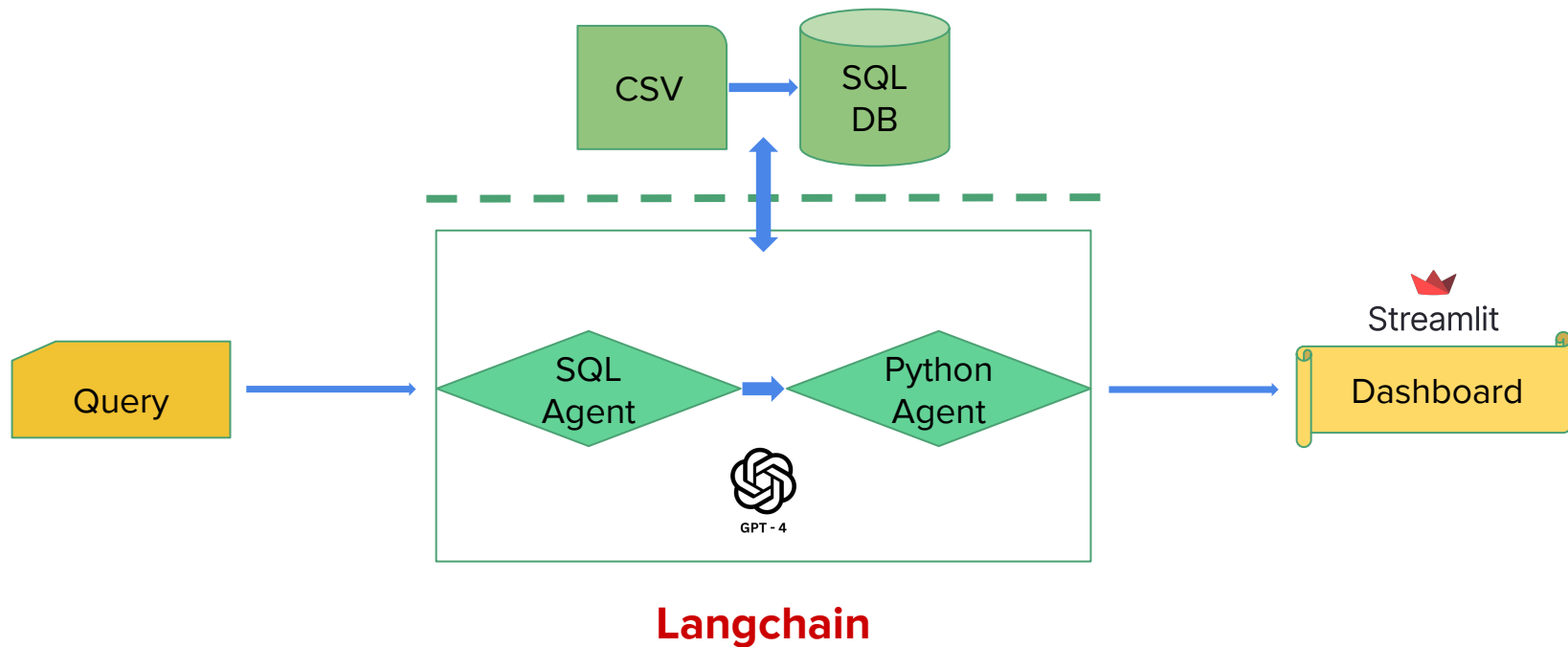
LangChain is an open source orchestration framework for the development of applications using large language models (LLMs).

Available in both Python- and Javascript-based libraries, LangChain's tools and APIs simplify the process of building LLM-driven applications like chatbots and virtual agents.



LangChain

Solution Approach



Tools and Tech

- **CSV** - Raw Data
- **SQL** - Data warehouse and query
- **Python** - Coding language
- **GPT-4** - LLM
- **Langchain** - Orchestration framework
- **Streamlit** - Dashboard

Challenges with Query/Code Generation

- Ambiguity in requirements
- Complexity of queries
- Compatibility and portability
- Error handling and debugging
- Security concerns
- Performance optimization
- Maintainability and scalability